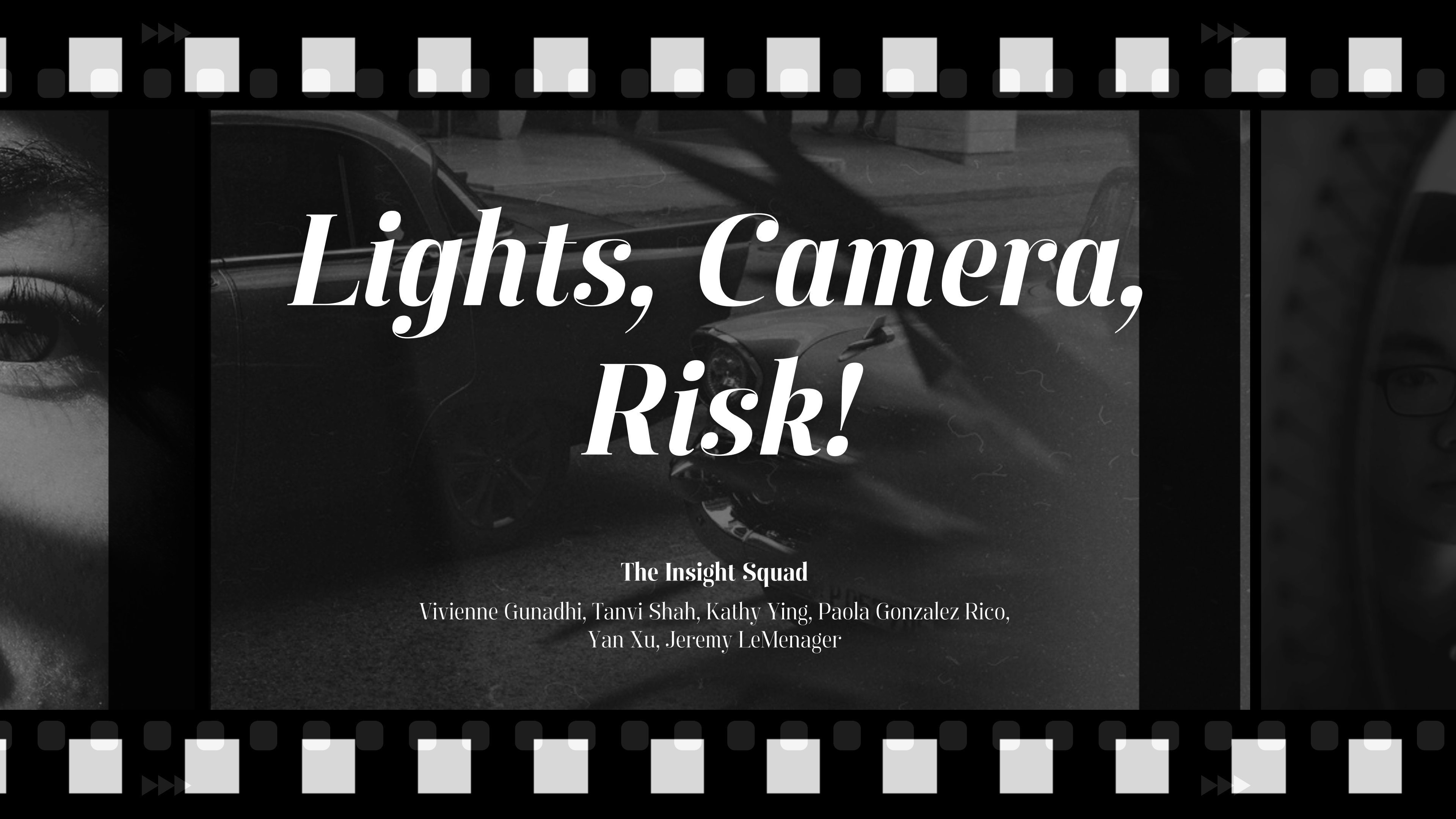




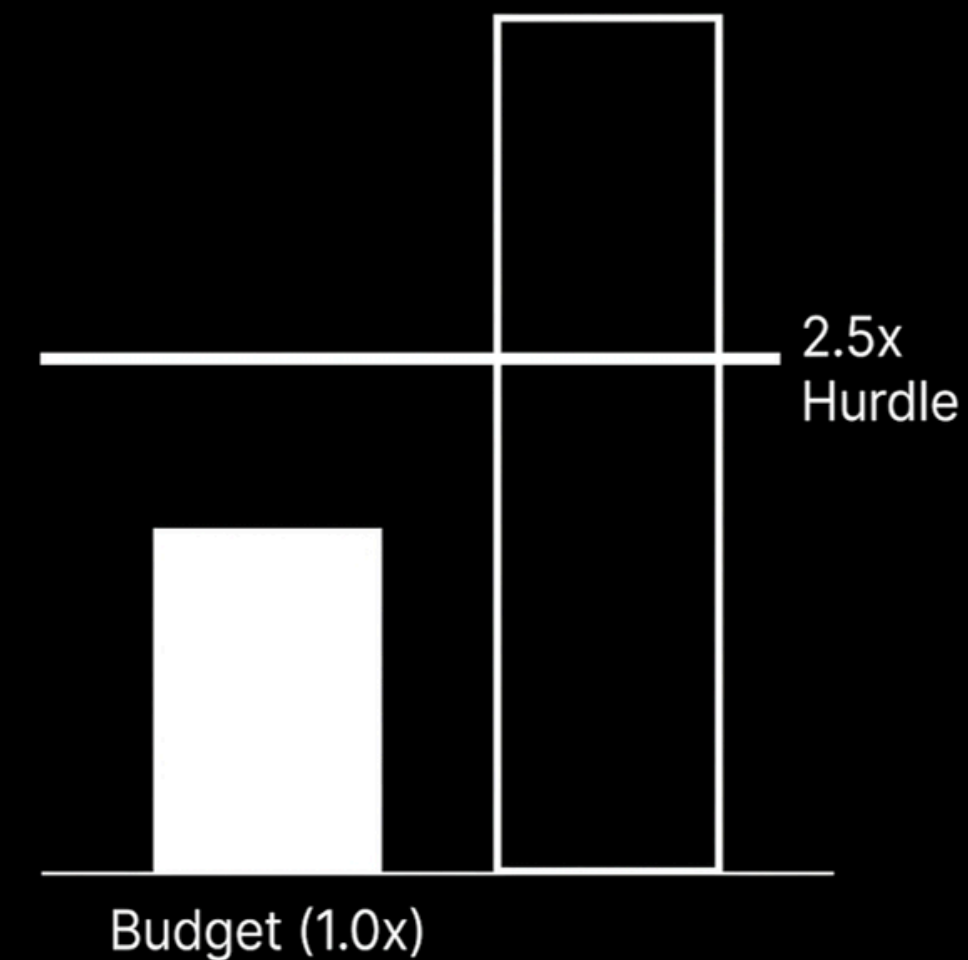
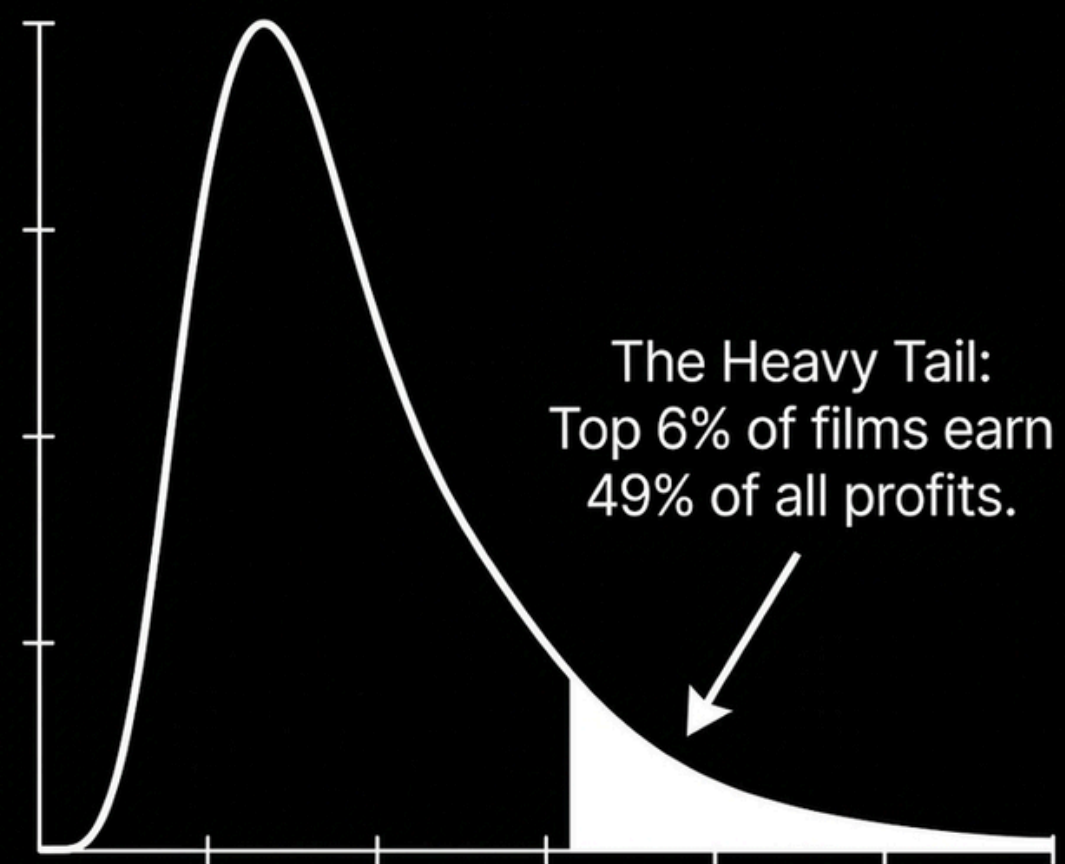
Lights, Camera, Risk!

The Insight Squad

Vivienne Gunadhi, Tanvi Shah, Kathy Ying, Paola Gonzalez Rico,
Yan Xu, Jeremy LeMenager

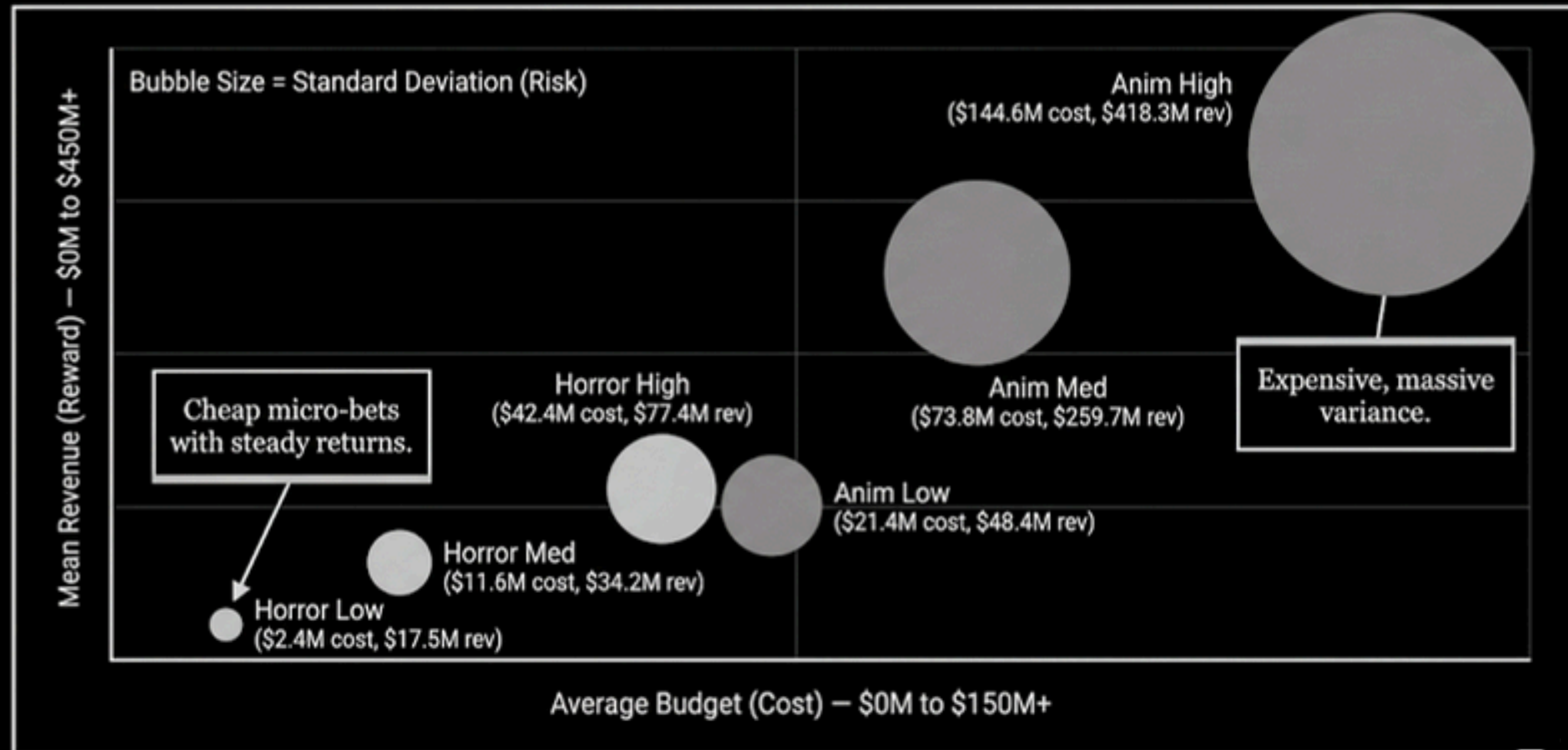
THE PROBLEM

49% of all films lose money



2.5x budget needed just
to break even.

The Investment Matrix: Mapping Cost, Reward, and Risk



Distributions fitted using AIC minimization via @RISK. LogNormal / Exponential models applied to genre-tier averages.

HISTORICAL DATA ANALYSIS

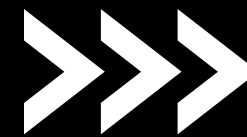
Historical Data

804 Historical Films

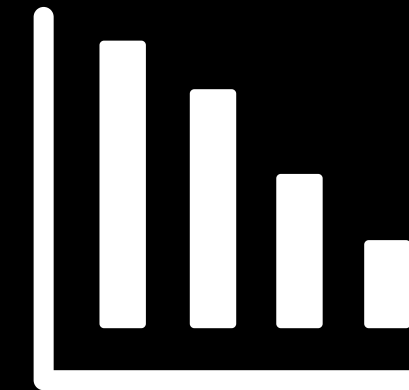
Budget tiers: Low, Med,
High

DECADE WEIGHTS (Recency Bias)

1915–1979	0.2
1980–1989	0.4
1990–1999	0.6
2000–2009	0.8
2010+	1



@Risk Distribution Fit



Fit lognormal
distributions for revenue
per genre
and budget tier

TWO-STAGE STOCHASTIC OPTIMIZATION



Solver: @Risk Optimizer (500 trials x 1,000 Montecarlo Simulations)

Uncertainties:

Film Revenue: R_{ij} for $i \in \{\text{Horror, Animation}\}$, $j \in \{\text{Low, Medium, High}\}$

Budget estimate: B_{ij} = Average of Historical data

Decision Variables:

Number of Films: N_{ij} for $i \in \{\text{Horror, Animation}\}$, $j \in \{\text{Low, Medium, High}\}$
 N_{total}

Budget for the next year: B_{S1}

Objective: max 50th percentile of $\{(\sum_{ij} N_{ij} \cdot R_{ij}) - (\sum_{ij} N_{ij} \cdot B_{ij})\}$

Constraints:

Budget: $\sum_{ij} N_{ij} \cdot B_{ij} \leq \300M

Genre: Horror share $\leq 68.85\%$ of total films (Mirror historical data)

Integer: $N_{ij} \in \mathbb{Z}_{\geq 0}$

Non-negativity: $N_{ij} \geq 0$

TWO-STAGE STOCHASTIC OPTIMIZATION



Solver: @Risk Optimizer (500 trials x 1,000 Montecarlo Simulations)

Uncertainties:

Film Revenue: R_{ij} for $i \in \{\text{Horror, Animation}\}$, $j \in \{\text{Low, Medium, High}\}$

Budget Estimate: B_{ij} = Average of Historical data

CVaR Estimate: $\lambda \times 1.65 \times \text{Portfolio}_{\text{stdev}}$

Cannibalization Estimate: $0.05 \times (x_{ij} - 5)$ for $x_{ij} > 5$

Decision Variables:

Final number of Films: x_{ij} for $i \in \{\text{Horror, Animation}\}$, $j \in \{\text{Low, Medium, High}\}$

Objective: $\max 50^{\text{th}}$ percentile of $\{\sum_{ij} x_{ij} \cdot R_{ij} - \sum_{ij} \text{Cannibalization}(x_{ij}) - (\sum_{ij} x_{ij} \cdot B_{ij}) - (\lambda \cdot 1.65 \times \text{Portfolio}_{\text{stdev}})\}$

Constraints:

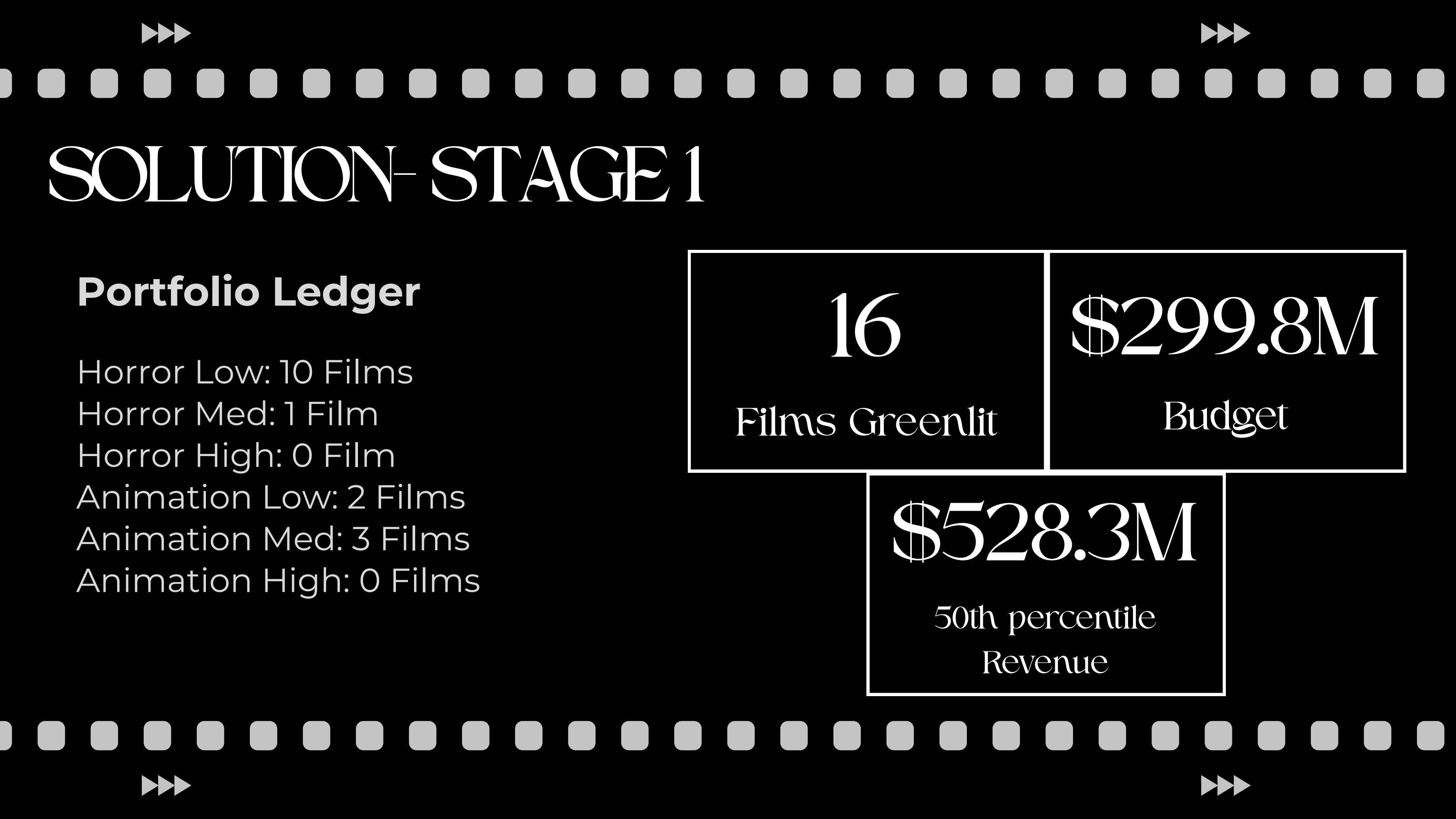
Budget: $\sum_{ij} x_{ij} \cdot B_{ij} \leq B_{S1}$

Genre: Animation share > 0

Total: $\sum_{ij} x_{ij} \leq N_{\text{total}}$

Integer: $x_{ij} \in \mathbb{Z}_{\geq 0}$

Non-negativity: $x_{ij} \geq 0$



SOLUTION- STAGE 1

Portfolio Ledger

Horror Low: 10 Films

Horror Med: 1 Film

Horror High: 0 Film

Animation Low: 2 Films

Animation Med: 3 Films

Animation High: 0 Films

16

Films Greenlit

\$299.8M

Budget

\$528.3M

50th percentile
Revenue

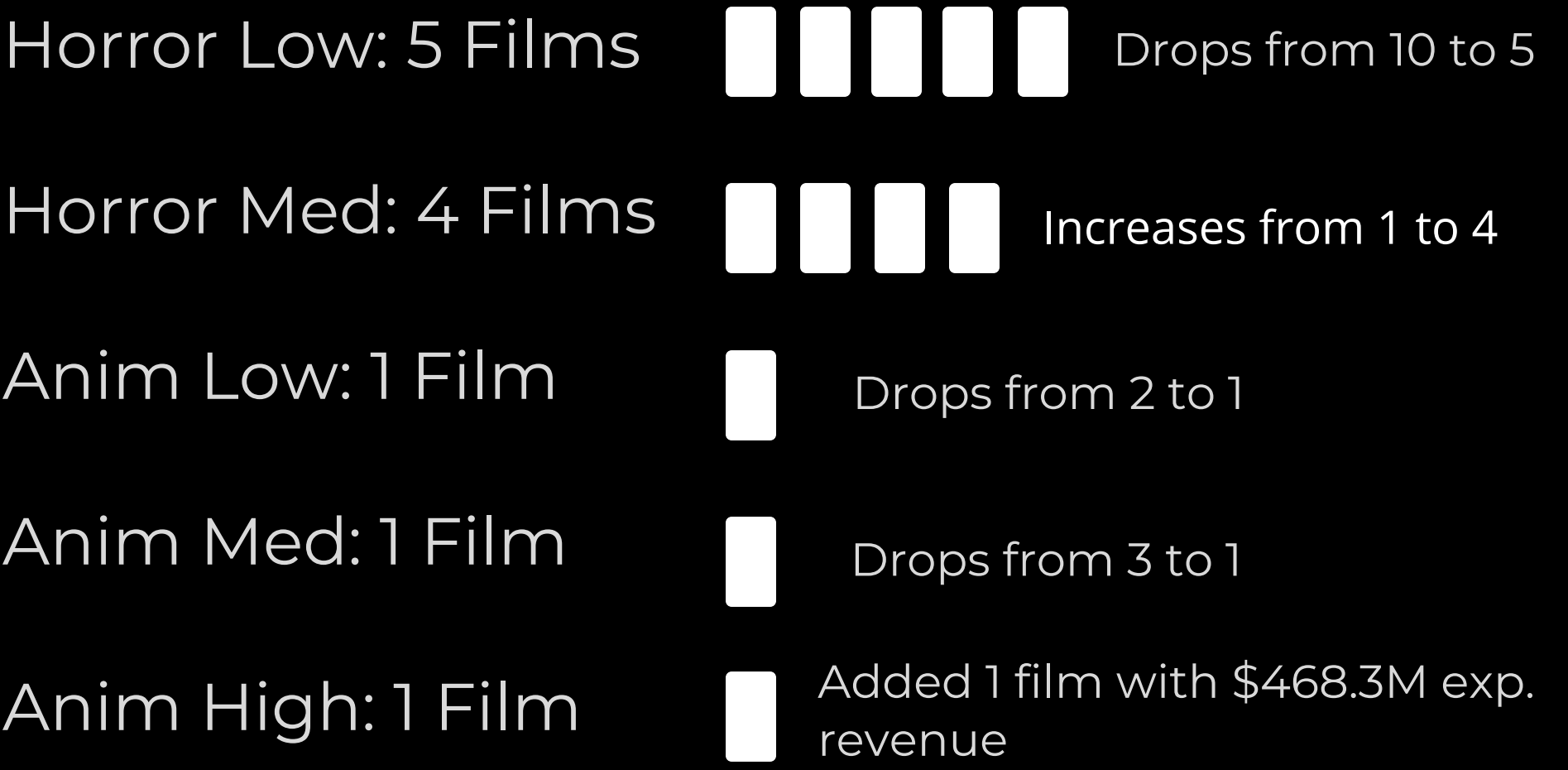
SOLUTION-STAGE 2

Stage 1 : 16 Films

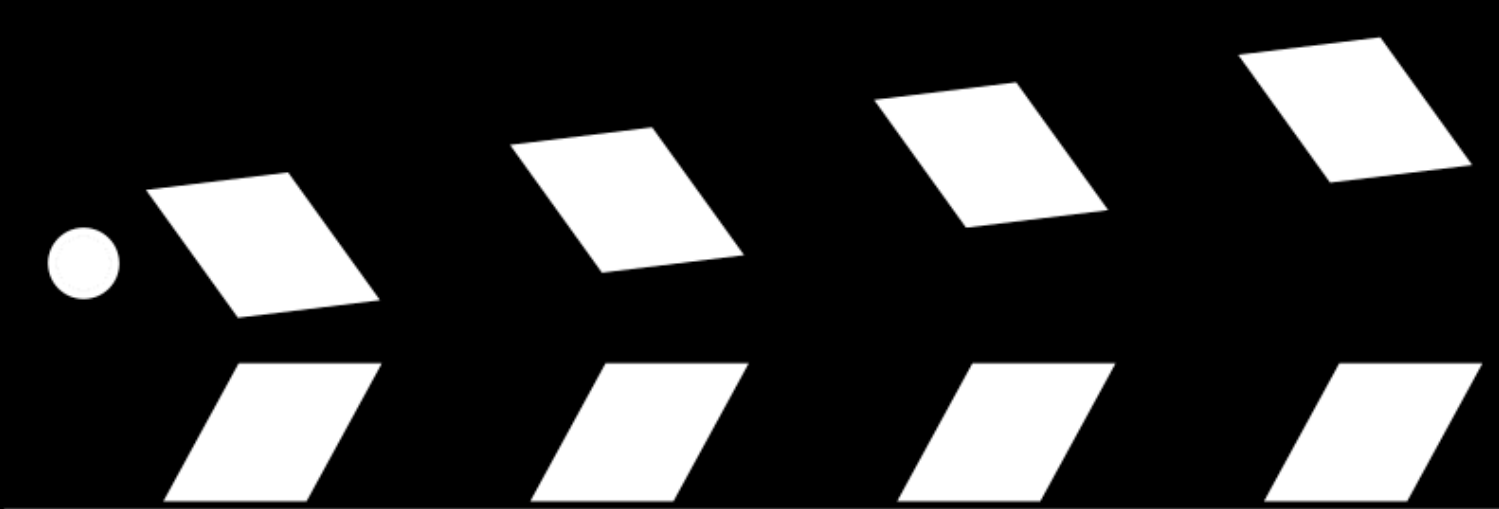


Cannibalization reduced Horror volume, freeing up budget for Animation.

Stage 2 : 12 Films



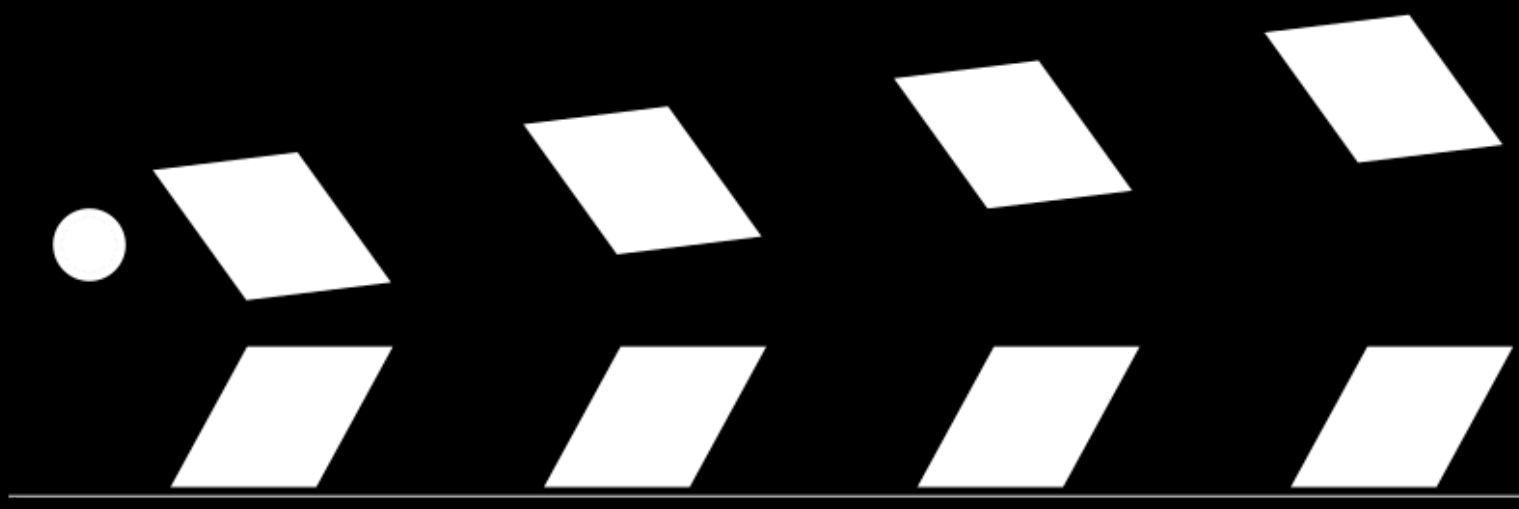
RECOMMENDATIONS



Adopt the 12-Film Annual Portfolio

Prioritize Animation Despite High Cost

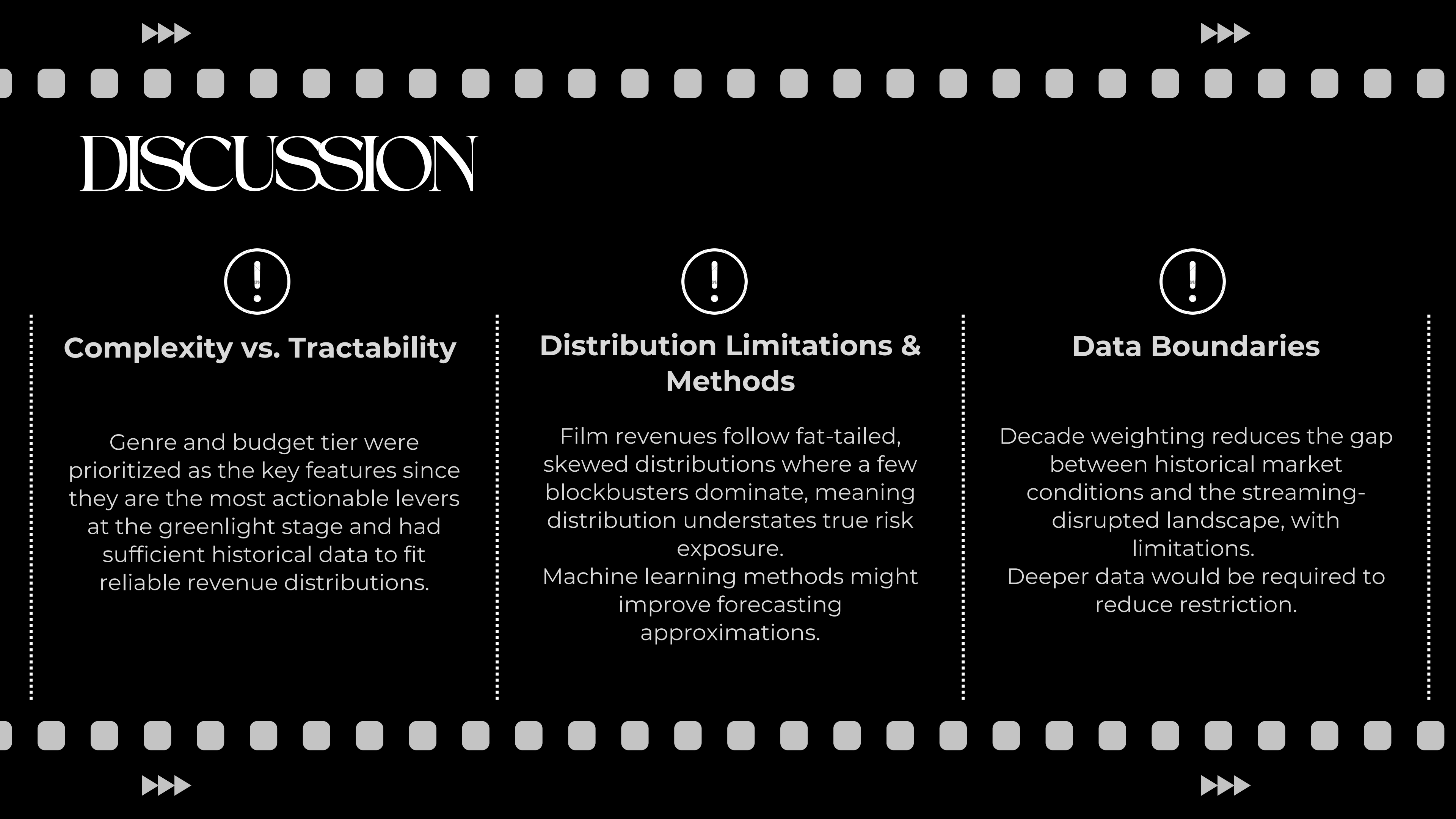
Cap Horror Low at 5 Films Per Year



Use Horror as a Portfolio Stabilizer

Recalibrate CVaR λ Annually

Wait-and-See: Contingent A&M Spending



DISCUSSION



Complexity vs. Tractability

Genre and budget tier were prioritized as the key features since they are the most actionable levers at the greenlight stage and had sufficient historical data to fit reliable revenue distributions.



Distribution Limitations & Methods

Film revenues follow fat-tailed, skewed distributions where a few blockbusters dominate, meaning distribution understates true risk exposure.
Machine learning methods might improve forecasting approximations.



Data Boundaries

Decade weighting reduces the gap between historical market conditions and the streaming-disrupted landscape, with limitations.
Deeper data would be required to reduce restriction.

FUTURE WORK

Dynamic CVaR Integration

A meaningful improvement would embed CVaR dynamically into the objective function, so that every simulation, contributes to the optimization and forces the solver to find allocations that are robust across the full distribution

Addressing the 23.5% Feasibility Rate

Future work should empirically validate this constraint and consider replacing it with a soft penalty to avoid widespread infeasibility.

Budget Uncertainty

Modeling the budget as a second stochastic variable would yield more conservative and realistic net revenue estimates, and interact with the CVaR mechanism to more accurately capture downside scenarios.

Expanded Genre Coverage and Cross-Genre Correlations

Extending to Action, Comedy, Drama, and Sci-Fi would allow the optimizer to exploit cross-genre diversification benefits



THANK *you*

